



1. Muestre que:

- $\sin(\theta) \sin(\varphi) = \frac{1}{2}(\cos(\theta - \varphi) - \cos(\theta + \varphi)).$
- $\cos(\theta) \cos(\varphi) = \frac{1}{2}(\cos(\theta - \varphi) + \cos(\theta + \varphi)).$
- $\sin(\theta) \cos(\varphi) = \frac{1}{2}(\sin(\theta - \varphi) + \sin(\theta + \varphi)).$

2. Muestre que:

- $\cos\left(\frac{n\pi(x+L)}{L}\right) = (-1)^n \cos\left(\frac{n\pi x}{L}\right).$
- $\sin\left(\frac{n\pi(x+L)}{L}\right) = (-1)^n \sin\left(\frac{n\pi x}{L}\right).$

3. Calcule el desarrollo de Fourier de la función  $x^2$  en  $(0, 2\pi)$ .

4. Calcule el desarrollo en base de senos de la función  $x(1-x)$  en  $[0, 1]$ .

5. Calcule el desarrollo en base de cosenos de la función  $f(x) = 2\pi x - \sin(2\pi x)$  en  $[0, 1]$ .

6. Calcule la solución del siguiente problema mixto para la ecuación de ondas

$$\begin{cases} u_{tt} - 4u_{xx} = 0 & \text{en } [0, \infty) \times [0, 1] \\ u(0, x) = \varphi(x) & \text{en } [0, 1] \\ u_t(0, x) = \psi(x) & \text{en } [0, 1] \\ u(t, 0) = u(t, 1) = 0 & \text{en } [0, \infty), \end{cases}$$

en los siguientes casos:

- a)  $\varphi(x) = \sin(\pi x) + \sin(2\pi x), \psi(x) = -\sin(\pi x) + \sin(2\pi x),$
- b)  $\varphi(x) = 0, \psi(x) = x(1-x).$

7. Calcule la solución del siguiente problema mixto para la ecuación de ondas:

$$\begin{cases} u_{tt} - 4u_{xx} = 0 & \text{en } [0, \infty) \times [0, 1] \\ u(0, x) = \varphi(x) & \text{en } [0, 1] \\ u_t(0, x) = \psi(x) & \text{en } [0, 1] \\ u_x(t, 0) = u_x(t, 1) = 0 & \text{en } [0, \infty), \end{cases}$$

en los siguientes casos:

- a)  $\varphi(x) = 1 + \cos(2\pi x), \psi(x) = 1 - \cos(\pi x),$
- b)  $\varphi(x) = \cos(2\pi x), \psi(x) = 2\pi x - \sin(2\pi x).$

8. (\*) Discuta la aplicabilidad del método de separación de variables a un problema del siguiente tipo:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{en } [0, \infty) \times [0, 1] \\ u(0, x) = \varphi(x) & \text{en } [0, 1] \\ u_t(0, x) = \psi(x) & \text{en } [0, 1] \\ u(t, 0) = u_x(t, 1) = 0 & \text{en } [0, \infty) . \end{cases}$$

9. Se considera el problema de Cauchy para la ecuación de ondas con fuerza externa dada por  $f$ :

$$\begin{cases} u_{tt} - c^2 u_{xx} = f(t, x) & \text{en } [0, \infty) \times \mathbb{R} \\ u(0, x) = \varphi(x) & \text{en } \mathbb{R} \\ u_t(0, x) = \psi(x) & \text{en } \mathbb{R} \end{cases} .$$

Muestre que la siguiente expresión proporciona una solución a este problema:

$$u(t, x) = \frac{1}{2} (\varphi(x + ct) + \varphi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(\sigma) d\sigma + \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(s, \sigma) d\sigma ds .$$

10. Se considera el siguiente problema de Cauchy para la ecuación de ondas:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{en } [0, \infty) \times \mathbb{R} \\ u(0, x) = \chi_{[-1, 1]}(x) & \text{en } \mathbb{R} \\ u_t(0, x) = 0 & \text{en } \mathbb{R} \end{cases} .$$

Calcule su solución mediante la correspondiente fórmula de d'Alembert. Repita el ejercicio con  $u(0, x) = (1 - |x|)\chi_{[-1, 1]}(x)$ .

11. Se considera el siguiente problema de Cauchy para la ecuación de ondas:

$$\begin{cases} u_{tt} - c^2 u_{xx} = \chi_{[-1, 1]}(x) & \text{en } [0, \infty) \times \mathbb{R} \\ u(0, x) = 0 & \text{en } \mathbb{R} \\ u_t(0, x) = 0 & \text{en } \mathbb{R} \end{cases} .$$

Calcule su solución mediante la correspondiente fórmula de d'Alembert.

12. Consideramos los siguientes problemas para la ecuación de ondas:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{en } [0, \infty) \times [0, \pi] \\ u(0, x) = \varphi(x) & \text{en } [0, \pi] \\ u_t(0, x) = \psi(x) & \text{en } [0, \pi] \\ u(t, 0) = u(t, \pi) = 0 & \text{en } [0, \infty) \end{cases} \quad (1)$$

$$\begin{cases} v_{tt} - c^2 v_{xx} = 0 & \text{en } [0, \infty) \times \mathbb{R} \\ v(0, x) = F(x) & \text{en } \mathbb{R} \\ v_t(0, x) = G(x) & \text{en } \mathbb{R} \end{cases} \quad (2)$$

donde  $F$  y  $G$  son las extensiones  $2\pi$ -periódicas a toda la recta de las extensiones impares de  $\varphi$  y  $\psi$  respectivamente, que supondremos suficientemente regulares. Muestre que la restricción de  $v(t, x)$  (que puede calcular mediante la fórmula de d'Alembert para (2)) al intervalo  $[0, \pi]$  es la solución  $u(t, x)$  de (1).

3. Calcule el desarrollo de Fourier de la función  $x^2$  en  $(0, 2\pi)$ .

$$f(x) = x^2 = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi x}{2\pi}\right) + b_n \sin\left(\frac{2n\pi x}{2\pi}\right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

$$a_0 = \frac{2}{2\pi} \int_0^{2\pi} x^2 dx = \frac{8\pi^2}{3}$$

$$a_n = \frac{2}{2\pi} \int_0^{2\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos(nx) dx = \left[ \begin{array}{l} u = x^2 \Rightarrow du = 2x dx \\ v = \int \cos(nx) dx = \frac{1}{n} \sin(nx) \end{array} \right] =$$

$$\frac{1}{\pi} \left( \left[ \frac{x^2}{n} \sin(nx) \right]_0^{2\pi} - \frac{2}{n} \int_0^{2\pi} x \sin(nx) dx \right) = \left[ \begin{array}{l} u = x \Rightarrow du = dx \\ v = \int \sin(nx) dx = -\frac{1}{n} \cos(nx) \end{array} \right] =$$

$$-\frac{2}{n\pi} \left( \left[ \frac{x}{n} \cos(nx) \right]_0^{2\pi} + \frac{1}{n} \int_0^{2\pi} \cos(nx) dx \right) = \frac{4\pi \cos(2\pi)}{n\pi} + \left[ \frac{1}{n^2} \sin(nx) \right]_0^{2\pi} = 1$$

$$b_n = \frac{2}{2\pi} \int_0^{2\pi} f(x) \sin(nx) dx = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin(nx) dx = \left[ \begin{array}{l} u = x^2 \Rightarrow du = 2x dx \\ v = \int \sin(nx) dx = -\frac{1}{n} \cos(nx) \end{array} \right] =$$

$$\frac{1}{\pi} \left( \left[ -\frac{x^2}{n} \cos(nx) \right]_0^{2\pi} + \frac{2}{n} \int_0^{2\pi} x \cos(nx) dx \right) = \left[ \begin{array}{l} u = x \Rightarrow du = dx \\ v = \int \cos(nx) dx = \frac{1}{n} \sin(nx) \end{array} \right] =$$

$$\frac{1}{\pi} \left( -\frac{4\pi^2}{n} + \frac{2}{n} \left( \left[ \frac{x}{n} \sin(nx) \right]_0^{2\pi} - \frac{1}{n} \int_0^{2\pi} \sin(nx) dx \right) \right) = \frac{-4\pi^2}{n} + \frac{2}{n^3} \left( \left[ \cos(nx) \right]_0^{2\pi} \right) = \frac{-4\pi^2}{n} = \frac{-4\pi}{n}$$

$$f(x) = x^2 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \cos(nx) - \frac{4\pi}{n} \sin(nx)$$

4. Calcule el desarrollo en base de senos de la función  $x(1-x)$  en  $[0, 1]$ .

$$f(x) = x(1-x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{1}\right) = \sum_{n=1}^{\infty} a_n \sin(n\pi x)$$

$$a_n = \frac{2}{1} \int_0^1 f(x) \sin(n\pi x) dx = 2 \int_0^1 x(1-x) \sin(n\pi x) dx = \left[ \begin{array}{l} u = x-x^2 \Rightarrow du = 1-2x dx \\ v = \int \sin(n\pi x) dx = -\frac{1}{n\pi} \cos(n\pi x) \end{array} \right] =$$

$$2 \left( \left[ -\frac{(x-x^2)}{n\pi} \cos(n\pi x) \right]_0^1 + \frac{1}{n\pi} \int_0^1 (1-2x) \cos(n\pi x) dx \right) = \left[ \begin{array}{l} u = 1-2x \Rightarrow du = -2 dx \\ v = \int \cos(n\pi x) dx = \frac{1}{n\pi} \sin(n\pi x) \end{array} \right]$$

$$= \frac{2}{n\pi} \left( \left[ \frac{(x-1)}{n\pi} \sin(n\pi x) \right]_0^1 + \frac{2}{n\pi} \int_0^1 \sin(n\pi x) dx \right) = \frac{-4}{n^3\pi^3} \left[ \cos(n\pi x) \right]_0^1 =$$

$$= \frac{-4}{n^3\pi^3} ((-1)^n - 1) = \begin{cases} 0 & n \text{ par} \\ \frac{8}{n^3\pi^3} & n \text{ impar} \end{cases}$$

$$f(x) = x(1-x) = \sum_{\substack{n=1 \\ n \text{ impar}}}^{\infty} \frac{8}{n^3\pi^3} \sin(n\pi x)$$

5. Calcule el desarrollo en base de cosenos de la función  $f(x) = 2\pi x - \sin(2\pi x)$  en  $[0, 1]$ .

$$f(x) = 2\pi x - \sin(2\pi x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{1}\right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x)$$

$$a_0 = \frac{2}{1} \int_0^1 f(x) dx = 2 \int_0^1 2\pi x - \sin(2\pi x) dx = 2\pi$$

$$a_n = \frac{2}{1} \int_0^1 f(x) \cos(n\pi x) dx = 2 \int_0^1 (2\pi x - \sin(2\pi x)) \cos(n\pi x) dx =$$

$$\begin{cases} 0 & n \text{ par} \\ 2 \left( \frac{-4}{n^2\pi} + \frac{4}{\pi(4^2-n^2)} \right) = \frac{8}{\pi} \left( \frac{-1}{n^2} + \frac{1}{(4^2-n^2)} \right) = \frac{8}{\pi} \frac{-1+4+n^2}{n^2(4^2-n^2)} = \frac{32}{\pi n^2(4^2-n^2)} & n \text{ impar} \end{cases}$$

$$* \int_0^1 2\pi x \cos(n\pi x) dx = \left[ \begin{matrix} u=x \Rightarrow du=dx \\ v=\sin(n\pi x) = \frac{1}{n\pi} \sin(n\pi x) \end{matrix} \right] = 2\pi \left( \frac{x}{n\pi} \sin(n\pi x) \right) \Big|_0^1 - \frac{1}{n\pi} \int_0^1 \sin(n\pi x) dx$$

$$= \frac{2}{n} \cdot \frac{1}{n\pi} [\cos(n\pi x)]_0^1 = \frac{2}{n^2\pi} ((-1)^n - 1) = \begin{cases} 0 & n \text{ par} \\ -\frac{4}{n^2\pi} & n \text{ impar} \end{cases}$$

$$* \int_0^1 -\sin(2\pi x) \cos(n\pi x) dx = -\frac{1}{2} \int_0^1 \sin(2\pi x + n\pi x) + \sin(2\pi x - n\pi x) dx =$$

$$-\frac{1}{2} \left( \int_0^1 \sin(\pi(4+2)x) + \sin(\pi(2-4)x) dx \right) =$$

$$-\frac{1}{2} \left( \frac{1}{\pi(4+2)} [\cos(\pi(4+2)x)]_0^1 - \frac{1}{\pi(2-4)} [\cos(\pi(2-4)x)]_0^1 \right) =$$

$$\frac{1}{2\pi} \left( \frac{1}{(4+2)} ((-1)^{4+2} - 1) + \frac{1}{(2-4)} ((-1)^{2-4} - 1) \right) =$$

$$\begin{cases} 0 & n \text{ par} \\ \frac{1}{2\pi} \left( \frac{-2}{4+2} - \frac{-2}{2-4} \right) = \frac{-1}{\pi} \left( \frac{1}{4+2} - \frac{1}{4-2} \right) = \frac{-1}{\pi} \left( \frac{4-2-4+2}{n^2-4} \right) = \frac{4}{\pi(4^2-n^2)} & n \text{ impar} \end{cases}$$

$$f(x) = \pi + \sum_{\substack{n \geq 1 \\ n \text{ impar}}} \frac{4}{\pi(4^2-n^2)} \cos(n\pi x)$$

6. Calcule la solución del siguiente problema mixto para la ecuación de ondas

$$\begin{cases} u_{tt} - 4u_{xx} = 0 & \text{en } [0, \infty) \times [0, 1] \\ u(0, x) = \varphi(x) & \text{en } [0, 1] \\ u_t(0, x) = \psi(x) & \text{en } [0, 1] \\ u(t, 0) = u(t, 1) = 0 & \text{en } [0, \infty), \end{cases}$$

en los siguientes casos:

Sabemos que la solución del problema mixto es única:

$$u(t, x) = \sum_{n=1}^{\infty} \left( \varphi_n \cos\left(\frac{n\pi c t}{L}\right) + \frac{\gamma_n c}{n\pi c} \sin\left(\frac{n\pi c t}{L}\right) \right) \sin\left(\frac{n\pi}{L} x\right), \quad c^2 = 4 \Rightarrow c = 2, \quad L = 1$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi x}{L}\right) \quad \varphi_n = \frac{2}{L} \int_0^L \varphi(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$\psi(x) = \sum_{n=1}^{\infty} \gamma_n \sin\left(\frac{n\pi x}{L}\right) \quad \gamma_n = \frac{2}{L} \int_0^L \psi(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$u(t, x) = \sum_{n=1}^{\infty} \left( \varphi_n \cos(2n\pi t) + \frac{\gamma_n}{2\pi} \sin(2n\pi t) \right) \sin(n\pi x)$$

$$u(0, x) = \varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin(n\pi x) \quad \varphi_n = 2 \int_0^1 \varphi(x) \sin(n\pi x) dx$$

$$u_t(0, x) = \psi(x) = \sum_{n=1}^{\infty} \gamma_n \sin(n\pi x) \quad \gamma_n = 2 \int_0^1 \psi(x) \sin(n\pi x) dx$$

a)  $\varphi(x) = \sin(\pi x) + \sin(2\pi x)$ ,  $\psi(x) = -\sin(\pi x) + \sin(2\pi x)$ ,

$$\varphi(x) = \sin(\pi x) + \sin(2\pi x) = \varphi_1 \sin(\pi x) + \varphi_2 \sin(2\pi x) + \sum_{n=3}^{\infty} \varphi_n \sin(n\pi x) \Rightarrow \varphi_1 = \varphi_2 = 1, \varphi_n = 0 \quad n \geq 3$$

$$\psi(x) = -\sin(\pi x) + \sin(2\pi x) = \gamma_1 \sin(\pi x) + \gamma_2 \sin(2\pi x) + \sum_{n=3}^{\infty} \gamma_n \sin(n\pi x) \Rightarrow \gamma_1 = -1, \gamma_2 = 1, \gamma_n = 0 \quad n \geq 3$$

Por tanto,  $u(t, x) = \left( \cos(2\pi t) - \frac{1}{2\pi} \sin(2\pi t) \right) \sin(\pi x) + \left( \cos(4\pi t) + \frac{1}{2\pi} \sin(4\pi t) \right) \sin(2\pi x)$

b)  $\varphi(x) = 0$ ,  $\psi(x) = x(1-x)$ .

•  $\varphi(x) = 0 = \sum_{n=1}^{\infty} \varphi_n \sin(n\pi x) \Rightarrow \varphi_n = 0 \quad \forall n \in \mathbb{N}$

•  $\psi(x) = x(1-x) = \sum_{n=1}^{\infty} \psi_n \sin(n\pi x)$ ,  $\psi_n = 2 \int_0^1 x(1-x) \sin(n\pi x) dx =$

$2 \int_0^1 x(1-x) \sin(n\pi x) dx = 2 \left[ -\int_0^1 x^2 \sin(n\pi x) dx + \int_0^1 x \sin(n\pi x) dx \right] =$

$2 \left( \frac{(-1)^n}{n\pi} + \frac{2}{n^3\pi^3} (1+(-1)^{n+1}) - \frac{(-1)^n}{n\pi} \right) = \frac{4}{n^3\pi^3} (1+(-1)^{n+1}) = \begin{cases} 0 & n \text{ par} \\ \frac{8}{n^3\pi^3} & n \text{ impar} \end{cases}$

\*  $\int_0^1 x \sin(n\pi x) dx = \left[ \begin{matrix} u=x \Rightarrow du=dx \\ v=\int \sin(n\pi x) dx = -\frac{1}{n\pi} \cos(n\pi x) \end{matrix} \right] = \left[ -\frac{x}{n\pi} \cos(n\pi x) \right]_0^1 + \frac{1}{n\pi} \int_0^1 \cos(n\pi x) dx$   
 $= -\frac{1}{n\pi} \cos(n\pi) + \frac{1}{(n\pi)^2} [\sin(n\pi x)]_0^1 = -\frac{1}{n\pi} \cos(n\pi) + \frac{1}{(n\pi)^2} \cancel{\sin(n\pi)}^0 = -\frac{(-1)^n}{n\pi}$

\*\*  $-\int_0^1 x^2 \sin(n\pi x) dx = \left[ \begin{matrix} u=x^2 \Rightarrow du=2x dx \\ v=\int \sin(n\pi x) dx = -\frac{1}{n\pi} \cos(n\pi x) \end{matrix} \right] =$

$\left[ -\frac{x^2}{n\pi} \cos(n\pi x) \right]_1^0 + \frac{2}{n\pi} \int_1^0 x \cos(n\pi x) dx = \left[ \begin{matrix} u=x \rightarrow du=dx \\ v=\int \cos(n\pi x) dx = \frac{1}{n\pi} \sin(n\pi x) \end{matrix} \right] =$

$\frac{1}{n\pi} \cos(n\pi) + \frac{2}{n\pi} \left[ \frac{x}{n\pi} \cancel{\sin(n\pi x)}^0 - \frac{1}{n\pi} \int_1^0 \sin(n\pi x) dx \right] =$

$\frac{1}{n\pi} \cos(n\pi) - \frac{2}{n^2\pi^2} \left[ -\frac{1}{n\pi} \cos(n\pi x) \right]_1^0 = \frac{(-1)^n}{n\pi} + \frac{2}{n^3\pi^3} (1 - \cos(n\pi)) = \frac{(-1)^n}{n\pi} + \frac{2}{n^3\pi^3} (1+(-1)^n)$

Por tanto,

$\varphi(x) = \frac{1}{2\pi} \sum_{n=1}^{\infty} \frac{8}{n^3\pi^3} \sin(2n\pi x) \sin(n\pi x) = \frac{4}{\pi^4} \sum_{\substack{n=1 \\ n \text{ impar}}}^{\infty} \frac{\sin(2n\pi x) \sin(n\pi x)}{n^3}$

7. Calcule la solución del siguiente problema mixto para la ecuación de ondas:

$$\begin{cases} u_{tt} - 4u_{xx} = 0 & \text{en } [0, \infty) \times [0, 1] \\ u(0, x) = \varphi(x) & \text{en } [0, 1] \\ u_t(0, x) = \psi(x) & \text{en } [0, 1] \\ u_x(t, 0) = u_x(t, 1) = 0 & \text{en } [0, \infty) \end{cases}$$

en los siguientes casos:

No conocemos la solución general para esas condiciones iniciales. Buscamos soluciones de la forma  $u(t, x) = T(t) X(x) \Rightarrow$

$$T'' X - 4T X'' = 0 \Rightarrow \frac{T''}{4T} = \frac{X''}{X} = -\lambda$$

$$\left. \begin{aligned} X'' + \lambda X &= 0 \\ X'(0) &= X'(1) = 0 \end{aligned} \right\}$$

$$\mu^2 + \lambda = 0 \Rightarrow \mu = \pm \sqrt{-\lambda}$$

$$\begin{aligned} * \lambda < 0 \Rightarrow X(x) &= C_1 e^{\sqrt{-\lambda} x} + C_2 e^{-\sqrt{-\lambda} x} \\ X'(x) &= \sqrt{-\lambda} C_1 e^{\sqrt{-\lambda} x} - \sqrt{-\lambda} C_2 e^{-\sqrt{-\lambda} x} \Rightarrow \end{aligned}$$

$$\left. \begin{aligned} X'(0) &= \sqrt{-\lambda} C_1 - \sqrt{-\lambda} C_2 = 0 \Rightarrow C_1 - C_2 = 0 \\ X'(1) &= \sqrt{-\lambda} e^{\sqrt{-\lambda}} C_1 - \sqrt{-\lambda} e^{-\sqrt{-\lambda}} C_2 = 0 \end{aligned} \right\} \Rightarrow C_1 = C_2 = 0 \Rightarrow X(x) = 0$$

$$* \lambda = 0 \Rightarrow X(x) = C_1 + C_2 x \Rightarrow X'(0) = X'(1) = C_2 = 0 \Rightarrow X(x) = C_1 e^{i\pi x}$$

$$* \lambda > 0 \Rightarrow X(x) = C_1 \cos(\sqrt{\lambda} x) + C_2 \sin(\sqrt{\lambda} x).$$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda} x)$$

$$X'(0) = C_2 \sqrt{\lambda} = 0 \Rightarrow \begin{cases} C_2 = 0 \\ \lambda = 0 \text{ (ya conocido)} \end{cases}$$

$$X'(1) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda}) = 0 \Rightarrow \begin{cases} C_1 = 0 \\ \lambda = 0 \text{ (ya conocido)} \\ \sin(\sqrt{\lambda}) = 0 \Rightarrow \sqrt{\lambda} = n\pi \mid n \in \mathbb{Z} \Rightarrow \\ \lambda = (n\pi)^2 \mid n \in \mathbb{N}_0 \end{cases}$$

$$X(x) = a_n \cos(n\pi x)$$

\*) Tendremos sol. de la forma  $u_n(t, x) = T_n(t) X_n(x)$

$$T_n'' + 4\lambda_n T_n = 0$$

$$n=0 \Rightarrow T_0'' = 0 \Rightarrow T_0(t) = a_0 + b_0 t$$

$$n \in \mathbb{N} \Rightarrow \mu^2 + 4\lambda_n = 0 \Rightarrow \mu = \pm \sqrt{-4\lambda_n} \Rightarrow T_n(t) = a_n \cos(\sqrt{4\lambda_n} t) + b_n \sin(\sqrt{4\lambda_n} t) \\ = a_n \cos(2n\pi t) + b_n \sin(2n\pi t)$$

$$u_0(t, x) = a_0 + b_0 t$$

$$u_n(t, x) = a_n \cos(2n\pi t) + b_n \sin(2n\pi t)$$

$$u(t, x) = \frac{a_0 + b_0 t}{2} + \sum_{n=1}^{\infty} (a_n \cos(2n\pi t) + b_n \sin(2n\pi t)) \cos(n\pi x)$$

$$u(0, x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x)$$

$$u_t(0, x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} b_n \cdot 2n\pi \cos(n\pi x)$$

a)  $\varphi(x) = 1 + \cos(2\pi x)$ ,  $\psi(x) = 1 - \cos(\pi x)$ ,

$$u_0(x) = \frac{a_0}{2} + a_1 \cos(\pi x) + a_2 \cos(2\pi x) + \sum_{n=2}^{\infty} a_n \cos(4n\pi x) = 1 + \cos(2\pi x) = \varphi(x)$$

$$\Rightarrow a_0 = 1, a_1 = 0, a_2 = 1$$

$$u_1(x) = \frac{b_0}{2} + 2\pi b_1 \cos(\pi x) + \sum_{n=2}^{\infty} 2\pi b_n \cos(4n\pi x) = 1 - \cos(\pi x) = \psi(x)$$

$$\Rightarrow b_0 = 2, b_1 = -\frac{1}{2\pi}$$

$$u_1(x) = 1 + -\frac{1}{2\pi} \sin(2\pi +) \cos(\pi x) + \cos(4\pi +) \cos(2\pi x)$$

b)  $\varphi(x) = \cos(2\pi x)$ ,  $\psi(x) = 2\pi x - \sin(2\pi x)$ .

$$u_0(x) = \frac{a_0}{2} + a_1 \cos(\pi x) + a_2 \cos(2\pi x) + \sum_{n=2}^{\infty} a_n \cos(4n\pi x) = 1 + \cos(2\pi x) = \varphi(x) \Rightarrow$$

$$\frac{a_0}{2} = a_1 = 0, a_2 = 1$$

$$u_1(x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} b_n \cdot 2n\pi \cos(4n\pi x) = 2\pi x - \sin(2\pi x) = \psi(x)$$

$$b_0 = 2 \int_0^1 \psi(x) dx = 2 \left( 2\pi \left[ \frac{x^2}{2} \right]_0^1 - \frac{1}{2\pi} [\cos(2\pi x)]_0^1 \right) = 2\pi$$

$$b_n \cdot 2n\pi = 2 \int_0^1 \psi(x) \cos(4n\pi x) dx \Rightarrow b_n = \frac{1}{n\pi} \int_0^1 (2\pi x - \sin(2\pi x)) \cos(4n\pi x) dx =$$

$$\begin{cases} 0 & n \text{ par} \\ \frac{-16}{n^3 \pi^2 (4 - n^2)} & n \text{ impar} \end{cases}$$

$$\sin(A) \cos(B) = \frac{1}{2} (\sin(A+B) + \sin(A-B))$$

$$* \int_0^1 2\pi x \cos(4n\pi x) dx - \int_0^1 \sin(2\pi x) \cos(4n\pi x) dx$$

$$u_1(x) = \pi x + \cos(4\pi x) \cos(2\pi x) + \sum_{\substack{n=1 \\ n \text{ impar}}}^{\infty} \frac{-16}{n^3 \pi^2 (4 - n^2)} \sin(2n\pi x)$$



8. (\*) Discuta la aplicabilidad del método de separación de variables a un problema del siguiente tipo:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{en } [0, \infty) \times [0, 1] \\ u(0, x) = \varphi(x) & \text{en } [0, 1] \\ u_t(0, x) = \psi(x) & \text{en } [0, 1] \\ u(t, 0) = u_x(t, 1) = 0 & \text{en } [0, \infty) \end{cases}$$

Buscamos soluciones de la forma  $u(t, x) = X(x)T(t)$ .

$$X T'' - c^2 X'' T = 0 \Rightarrow \frac{X''}{X} = \frac{T''}{T} = -\lambda, \lambda \in \mathbb{R}$$

$$\rightarrow c^2 X'' + \lambda X = 0$$

$$c^2 \mu + \lambda = 0 \Rightarrow \mu = \pm \frac{1}{c} \sqrt{\lambda}$$

$$* \lambda = 0 \Rightarrow X(x) = c_1 + c_2 x, X'(x) = c_2$$

$$X(0) = c_1 = 0, X'(1) = c_2 = 0 \Rightarrow X(x) \equiv 0$$

$$* \lambda < 0 \Rightarrow X(x) = c_1 e^{\frac{1}{c} \sqrt{-\lambda} x} + c_2 e^{-\frac{1}{c} \sqrt{-\lambda} x}, X'(x) = \frac{c_1}{c} \sqrt{-\lambda} e^{\frac{1}{c} \sqrt{-\lambda} x} - \frac{c_2}{c} \sqrt{-\lambda} e^{-\frac{1}{c} \sqrt{-\lambda} x}$$

$$\begin{aligned} X(0) &= c_1 + c_2 = 0 \\ X'(1) &= \frac{c_1}{c} \sqrt{-\lambda} e^{\frac{1}{c} \sqrt{-\lambda}} - \frac{c_2}{c} \sqrt{-\lambda} e^{-\frac{1}{c} \sqrt{-\lambda}} = 0 \end{aligned} \left. \begin{aligned} & \\ & \end{aligned} \right\} c_1 = c_2 = 0 \Rightarrow X(x) \equiv 0$$

$$* \lambda > 0 \Rightarrow X(x) = c_1 \cos\left(\frac{1}{c} \sqrt{\lambda} x\right) + c_2 \sin\left(\frac{1}{c} \sqrt{\lambda} x\right)$$

$$X'(x) = -\frac{c_1}{c} \sqrt{\lambda} \sin\left(\frac{1}{c} \sqrt{\lambda} x\right) + \frac{c_2}{c} \sqrt{\lambda} \cos\left(\frac{1}{c} \sqrt{\lambda} x\right)$$

$$X(0) = c_1 = 0$$

$$X'(1) = \frac{c_2}{c} \sqrt{\lambda} \cos\left(\frac{1}{c} \sqrt{\lambda}\right) = 0 \Rightarrow \begin{cases} c_2 = 0 \Rightarrow X(x) \equiv 0 \\ \lambda = 0 \Rightarrow \text{ya considerado} \\ \cos\left(\frac{1}{c} \sqrt{\lambda}\right) = 0 \Rightarrow \frac{\sqrt{\lambda}}{c} = \frac{\pi}{2} + \pi k \end{cases}$$

$$\text{Sin embargo, } 0 < \frac{\sqrt{\lambda}}{c} < \frac{\pi}{2} + \pi k \Rightarrow k > -\frac{1}{2} \Rightarrow k \in \mathbb{N}_0 \Rightarrow \lambda = c^2 \left(\frac{\pi}{2} + \pi k\right)^2, k \in \mathbb{Z}$$

$$X_k(x) = \sin\left(\frac{1}{c} c \left(\frac{\pi}{2} + \pi k\right) x\right) = \sin\left(\left(\frac{\pi}{2} + \pi k\right) x\right) \quad c_2 \in \mathbb{R}, k \in \mathbb{N}_0$$

$$T_k'' + \lambda_k T_k = 0$$

$$\mu \in \mathbb{N} \Rightarrow \mu^2 + \lambda_k = 0 \Rightarrow \mu^2 = -\lambda_k = \pm \sqrt{\lambda_k} i \Rightarrow$$

$$T_k(t) = a_k \cos(c(\frac{\pi}{2} + \pi k) t) + b_k \sin(c(\frac{\pi}{2} + \pi k) t)$$

$$u_k = \left[ a_k \cos(c(\frac{\pi}{2} + \pi k)) + b_k \sin(c(\frac{\pi}{2} + \pi k)) \right] \sin\left(\left(\frac{\pi}{2} + \pi k\right) x\right)$$

$$u(t, x) = \sum_{k=0}^{\infty} \left[ a_k \cos(c(\frac{\pi}{2} + \pi k)) + b_k \sin(c(\frac{\pi}{2} + \pi k)) \right] \sin\left(\left(\frac{\pi}{2} + \pi k\right) x\right)$$

10. Se considera el siguiente problema de Cauchy para la ecuación de ondas:

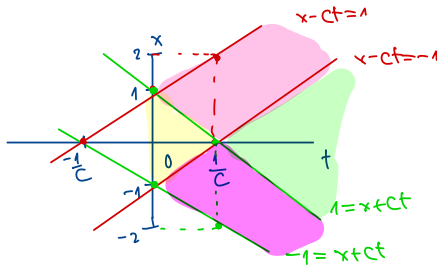
$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{en } [0, \infty) \times \mathbb{R} \\ u(0, x) = \chi_{[-1, 1]}(x) & \text{en } \mathbb{R} \\ u_t(0, x) = 0 & \text{en } \mathbb{R} \end{cases}.$$

Calcule su solución mediante la correspondiente fórmula de d'Alembert. Repita el ejercicio con

$$u(0, x) = (1 - |x|)\chi_{[-1, 1]}(x).$$

Sabemos que la solución es dada por

$$u(t, x) = \frac{\chi_{[-1, 1]}(x+ct) + \chi_{[-1, 1]}(x-ct)}{2} + \int_{x-ct}^{x+ct} 0(s) ds = \begin{cases} \frac{1+1}{2} = 1 & x \in [-1+ct, 1-ct] \\ \frac{1+0}{2} = \frac{1}{2} & x \in [-1+ct, 1+ct] \cup [-1-ct, 1-ct] \\ 0 & \text{en otro caso} \end{cases}$$



$$u(0, x) = (1 - |x|)\chi_{[-1, 1]}(x) \Rightarrow u(t, x) = \begin{cases} \frac{|x+ct| \cdot 1 + |x+ct| \cdot 1}{2} = |x+ct| & x \in [-1+ct, 1-ct] \\ \frac{0 + |x-ct| \cdot 1}{2} = \frac{|x-ct|}{2} & x \in [1+ct, 1+ct] \\ \frac{|x+ct| \cdot 1 + 0}{2} = \frac{|x+ct|}{2} & x \in [-1-ct, 1-ct] \\ 0 & \text{en otro caso} \end{cases}$$

11. Se considera el siguiente problema de Cauchy para la ecuación de ondas:

$$\begin{cases} u_{tt} - c^2 u_{xx} = \chi_{[-1, 1]}(x) & \text{en } [0, \infty) \times \mathbb{R} \\ u(0, x) = 0 & \text{en } \mathbb{R} \\ u_t(0, x) = 0 & \text{en } \mathbb{R} \end{cases}.$$

Calcule su solución mediante la correspondiente fórmula de d'Alembert.